

Research Report: NV-diamond use-cases for ECG detection

Topic

NV-diamond use-cases for ECG detection

Search Settings

- Requested papers: 3
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- Sort mode: Relevance
- Date filter: All publication dates
- LLM model: gpt-5.4-mini

Topic Understanding

The topic is interpreted as a research search about NV-diamond use-cases for ECG detection, expanded with terminology from web snippets and a fixed domain glossary.

Expanded Keywords

- Primary Terms: NV diamond, nitrogen vacancy diamond, nitrogen-vacancy center, ECG, electrocardiography, magnetocardiography
- Related Terms: electrocardiography, magnetocardiography, cardiac monitoring, NV diamond, nitrogen vacancy diamond, nitrogen-vacancy center, nv, diamond
- Domain Terms: quantum sensing, biomedical sensing, magnetometry, cardiac monitoring
- Synonyms: NV diamond, nitrogen vacancy diamond, nitrogen-vacancy center, ECG, electrocardiography, magnetocardiography, cardiac monitoring, diamond quantum sensor
- Research Keywords: nv, diamond, centre, detection, quantum, applications
- Web Context:
 - Understanding NV Diamond Technology: Principles & Applications ? Adámas ...: The nitrogen-vacancy (NV) center in diamond is used for detection of local magnetic or electric fields and/or temperature. Depending on use case and required sensitivity, different designs of diamond plates containing NV centers can be produced.
 - PDF Diamonds for Medical Applications - University of Oxford: We are able to use diamond for quantum applications because of a particular colour centre called the nitrogen-vacancy (NV) centre. An NV centre is formed by a nitrogen atom replacing a carbon atom, adjacent to a missing carbon atom (the 'vacancy'). The NV centre emits red light when we shine green light on it and we can detect this red florescence from a single centre. These colour centres ...
 - arXiv:2405.02376v1 [physics.med-ph] 3 May 2024: rged Nitrogen-Vacancy (NV) centers in diamond . The magnetic cardiac signal of a living rat, characterized by an approximately 20 pT amplitude in the R-wave, is successfully captured through non-invasive measurement using this innovative solid-state spin sensor. To detect these extremely weak biomag-netic signals, we utilize sensitivity-enhancing

Generated Search Query

(all:"NV diamond" OR all:"nitrogen vacancy diamond" OR all:"nitrogen-vacancy center" OR all:"nv" OR all:"diamond") AND (all:"ECG" OR all:"electrocardiography" OR all:"magnetocardiography" OR all:"cardiac monitoring" OR all:"quantum sensing" OR all:"biomedical sensing" OR all:"magnetometry" OR all:"diamond quantum sensor")

Overview

This report summarizes 3 relevant arXiv paper(s) for NV-diamond use-cases for ECG detection. The retrieved abstracts suggest the following research context: Nitrogen-vacancy (NV) spin ensembles in diamond provide an advanced magnetic sensing platform, with applications in both the physical and life sciences. The development of isotopically engineered ^{15}N NV diamond offers advantages over naturally occurring ^{14}N NV for magnetometry, due to its simpler hyperfine structure. However, for sensing modalities requiring a bias magnetic field not aligned with the sensing NV axis, the absence of a quadrupole moment in the ^{15}N nuclear spin leads to pronounced envelope modulation effects in time-dependent measurements of ^{15}N NV spin evolution. While such behavior in spin echo experiments are well studied, analogous effects in Ramsey measurements and the implications for magnetometry remain under-explored. Here, we derive the modulated ^{15}N NV Ramsey response to a misaligned bias field, using a simple vector description of the effective m...

Research Summary

Across the retrieved papers, common themes include threshold, spin, sensitivity, diamond, magnetic, laser. The abstracts indicate a cluster of work around methods, evaluation, system design, and practical limitations. This summary is generated from repeated terms and leading abstract sentences rather than an external LLM.

Key Papers

1. Ramsey Envelope Modulation in NV Diamond Magnetometry (2022-05-05)

- Authors: Jner Tzern Oon, Jiashen Tang, Connor A. Hart, Kevin S. Olsson, Matthew J. Turner, Jennifer M. Schloss, Ronald L. Walsworth

- Abstract: Nitrogen-vacancy (NV) spin ensembles in diamond provide an advanced magnetic sensing platform, with applications in both the physical and life sciences. The development of isotopically engineered ^{15}N NV diamond offers advantages over naturally occurring ^{14}N NV for magnetometry, due to its simpler hyperfine structure. However, for sensing...

- URL: <http://arxiv.org/abs/2205.02387v1>

2. A robust laser cavity platform for NV-diamond singlet infrared absorption magnetometry (2026-04-21)

- Authors: Shao Qi Lim, Alexander A. Wood, Brett C. Johnson, Qiang Sun, Jan Jeske, Hiroshi Abe, Takeshi Ohshima, David J. Ottaway, Heike Ebendorff-Heidepriem, Robert E. Scholten, Andrew D. Greentree, Brant C. Gibson

- Abstract: The negatively charged nitrogen-vacancy center (NV^-) in diamond is a versatile platform for quantum magnetometry under ambient conditions. Recently, laser threshold magnetometry (LTM) has been proposed as a means to significantly enhance the sensitivity of NV-based magnetometers by incorporating a diamond hosting NV^- centers within a laser...

- URL: <http://arxiv.org/abs/2604.18937v1>

3. Sensitivity Optimization for NV-Diamond Magnetometry (2019-03-19)

- Authors: John F. Barry, Jennifer M. Schloss, Erik Bauch, Matthew J. Turner, Connor A. Hart, Linh M. Pham, Ronald L. Walsworth

- Abstract: Solid-state spin systems including nitrogen-vacancy (NV) centers in diamond constitute an increasingly favored quantum sensing platform. However, present NV ensemble devices exhibit sensitivities orders of magnitude away from theoretical limits. The sensitivity shortfall both handicaps existing implementations and curtails the envisioned...

- URL: <http://arxiv.org/abs/1903.08176v2>

Main Findings

- Nitrogen-vacancy (NV) spin ensembles in diamond provide an advanced magnetic sensing platform, with applications in both the physical and life sciences.
- The negatively charged nitrogen-vacancy center (NV⁻) in diamond is a versatile platform for quantum magnetometry under ambient conditions.
- Solid-state spin systems including nitrogen-vacancy (NV) centers in diamond constitute an increasingly favored quantum sensing platform.

Research Gaps

- Cross-paper comparison may be limited because abstracts often emphasize each paper's method more than standardized benchmarks or negative results.
- The retrieved set may underrepresent adjacent work outside arXiv or papers that use different terminology.

Future Directions

- Explore deeper combinations of recurring themes such as threshold, spin, sensitivity, diamond.
- Compare the retrieved approaches on shared datasets and consistent evaluation metrics.
- Study robustness, reproducibility, and failure cases across realistic usage settings.
- Connect the strongest ideas from multiple papers into a unified experimental baseline.

References

- Jier Tzern Oon, Jiashen Tang, Connor A. Hart, Kevin S. Olsson, Matthew J. Turner, Jennifer M. Schloss, Ronald L. Walsworth. Ramsey Envelope Modulation in NV Diamond Magnetometry. arXiv, 2022-05-05. <http://arxiv.org/abs/2205.02387v1>
- Shao Qi Lim, Alexander A. Wood, Brett C. Johnson, Qiang Sun, Jan Jeske, Hiroshi Abe, Takeshi Ohshima, David J. Ottaway, Heike Ebendorff-Heidepriem, Robert E. Scholten, Andrew D. Greentree, Brant C. Gibson. A robust laser cavity platform for NV-diamond singlet infrared absorption magnetometry. arXiv, 2026-04-21. <http://arxiv.org/abs/2604.18937v1>
- John F. Barry, Jennifer M. Schloss, Erik Bauch, Matthew J. Turner, Connor A. Hart, Linh M. Pham, Ronald L. Walsworth. Sensitivity Optimization for NV-Diamond Magnetometry. arXiv, 2019-03-19. <http://arxiv.org/abs/1903.08176v2>